

An Empirical Analysis of Hallucination Behavior in LLMs in the Medical Domain

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Abstract

Large Language Models (LLMs) have demonstrated strong capabilities in generating fluent, contextually relevant text. However, they frequently produce hallucinations — outputs that appear plausible but are factually incorrect, unsupported, or fabricated. This problem is especially critical in high-stakes domains such as medicine, where incorrect information can have serious consequences. We present an empirical study of hallucination behavior across three model families (Llama-3-8B, BioMistral-7B, and BioGPT) on two medical QA benchmarks (MedQA and Pub-MedQA), evaluating four intervention conditions: zero-shot prompting, chain-of-thought (CoT) prompting, retrieval-augmented generation (RAG), and supervised fine-tuning (SFT). Our results show that larger instruction-tuned models benefit most from RAG, while the effect of CoT depends on the task and model: it hurts performance on MedQA with Llama-3-8B-Instruct but improves over zero-shot on Pub-MedQA with BioMistral-7B. Fine-tuning stabilizes BioGPT’s output by eliminating parse failures on MedQA, but does not materially improve accuracy. Combining fine-tuning with RAG, however, leads to catastrophic output degradation, suggesting that task-specific fine-tuning and retrieval augmentation are incompatible in their current forms for this model.

1 Introduction

Large Language Models (LLMs) have transformed natural language processing by generating coherent, relevant text. However, a major challenge is their propensity to produce hallucinations — outputs that may be fluent and plausible but are factually incorrect, unsupported by evidence, or entirely fabricated. This issue is particularly concerning in medicine, where incorrect information can lead to serious consequences for patient care and safety.

Despite the growing body of research on hallucination in general NLP contexts, relatively little

work systematically characterizes hallucination behavior in medical LLMs or directly compares the effectiveness of inference-time versus training-time mitigation strategies. In this paper, we address this gap by evaluating multiple LLMs on standardized medical QA benchmarks under four experimental conditions: zero-shot prompting, chain-of-thought prompting, retrieval-augmented generation, and supervised fine-tuning. By comparing these interventions, we ask not only how often LLMs hallucinate on medical questions, but which type of intervention works best — and whether combining them yields further gains.

2 Related Work

2.1 Hallucination in Large Language Models

Hallucination is a widely recognized problem in LLMs, where a model generates fluent and plausible text that is factually incorrect, unsupported by the input, or inconsistent with external knowledge. Prior work has categorized hallucinations into intrinsic hallucinations, which contradict the source context, and extrinsic hallucinations, which introduce information not supported by the provided evidence (Ji et al., 2023). Hallucination has also been studied in abstractive summarization, where models often generate coherent but unfaithful content (Maynez et al., 2020). These findings are relevant to medical LLMs because medical responses require both factual accuracy and faithfulness to clinical context.

2.2 Medical Question Answering and Clinical Reasoning

Recent work has shown that LLMs can achieve strong performance on medical QA benchmarks such as MedQA, MedMCQA, and PubMedQA, but remain vulnerable to confident errors on complex clinical reasoning tasks (Liévin et al., 2024). Medical benchmarks provide a useful setting for

studying hallucination because they offer verifiable ground-truth answers. Prior studies have also developed medical hallucination benchmarks, such as Med-HALT, which evaluates reasoning-based and memory-based hallucination in medical LLMs (Pal et al., 2023).

2.3 Chain-of-Thought Prompting

Chain-of-thought (CoT) prompting encourages a model to generate intermediate reasoning steps before producing a final answer (Wei et al., 2022). While CoT has been shown to improve reasoning on complex tasks, it can also produce longer generations that introduce additional opportunities for error. In medical QA, a model may generate a fluent but incorrect reasoning chain before giving its final answer, and CoT outputs may be harder to parse reliably (Wang et al., 2023).

2.4 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) combines a generative model with an external retrieval component, allowing the model to condition its output on retrieved documents rather than relying only on parametric knowledge (Lewis et al., 2020). In the medical domain, retrieval can help ground model outputs in biomedical abstracts or clinical documents, potentially reducing extrinsic hallucination. However, RAG does not fully eliminate hallucination, as a model may still misinterpret retrieved evidence or fail to align its answer with the retrieved context.

2.5 Fine-Tuning for Medical NLP

Supervised fine-tuning (SFT) on domain-specific data has been shown to improve model performance on medical tasks (Singhal et al., 2023). Domain-specific pretraining, as in BioGPT (Luo et al., 2022) and BioMistral (Labrak et al., 2024), has also been proposed as a way to reduce hallucination by internalizing medical knowledge during training. However, the relative effectiveness of fine-tuning versus inference-time interventions such as RAG remains underexplored, motivating our direct comparison.

3 Methods

3.1 Datasets

We evaluate on two medical QA benchmarks that represent different task types and hallucination detection modalities.

MedQA (USMLE) (Jin et al., 2021) consists of approximately 12,000 multiple-choice questions derived from United States Medical Licensing Examinations, covering a broad range of clinical topics. Gold-standard answer letters allow direct accuracy measurement. We evaluate on the held-out test split, sampling 450–500 items per experimental condition.

PubMedQA (Jin et al., 2019) consists of 450 expert-annotated yes/no/maybe questions derived from PubMed research abstracts. Each question is paired with the source abstract, making it ideal for evaluating context-grounded generation and retrieval augmentation. We use the full labeled training split (450 items) for all PubMedQA conditions.

3.2 Models

We compare three models spanning capability tiers:

Meta-Llama-3-8B-Instruct is a large general-purpose instruction-tuned model serving as our strong baseline. It was not fine-tuned on medical data.

BioMistral-7B (Labrak et al., 2024) is a medically-adapted model built on Mistral-7B and further pretrained on biomedical corpora. It allows us to test whether domain-specific pretraining reduces hallucination relative to a general model.

BioGPT (Luo et al., 2022) is a smaller (1.5B parameter) generative model pretrained on PubMed abstracts. It serves as both a small-model baseline and the target for our supervised fine-tuning experiments.

3.3 Experimental Conditions

We evaluate each model under the following conditions:

Zero-shot prompting provides the model with the question and answer choices (MedQA) or question alone (PubMedQA) with no additional context or reasoning instruction.

Chain-of-thought (CoT) prompting asks the model to reason step-by-step before providing a final answer, ending with Answer : <letter>.

Retrieval-augmented generation (RAG) uses BM25 retrieval over the PubMedQA abstract corpus with top- $k=2$ abstracts prepended as context, using a fixed prompt template across all models.

Supervised fine-tuning (SFT) fine-tunes BioGPT on the full MedQA training split (10,178 samples) for 3 epochs using standard next-token prediction loss, learning rate 2×10^{-5} , effective

batch size 16, and fp16 mixed precision on a single A100 GPU.

SFT + RAG applies RAG at inference time to the fine-tuned BioGPT model, testing whether the two interventions are complementary or redundant.

3.4 Evaluation Metrics

Accuracy is computed as the proportion of outputs whose extracted answer exactly matches the gold-standard answer. For MedQA, this means the extracted letter (A–D) matches the gold answer letter. For PubMedQA, this means the extracted decision (yes/no/maybe) matches the gold label.

Error rate is defined as $1 - \text{accuracy}$ over all outputs including parse failures. For MedQA conditions (zero-shot, CoT, SFT), we use error rate rather than “hallucination rate” since there is no retrieved context to assess faithfulness against — incorrect answers reflect factual errors rather than hallucination in the retrieval faithfulness sense.

Hallucination rate is reserved for RAG conditions on PubMedQA, where it captures outputs that are either incorrect or unfaithful to the retrieved context. Parse failures are included in the hallucination count since an unparseable output is clinically unusable.

Parse failure rate is reported separately as the proportion of outputs from which no structured answer could be extracted using regex pattern matching. For MedQA, the extractor looks for `Answer: [A–D]` or a leading letter. For PubMedQA, it looks for yes, no, or maybe at the start of or within the output. Parse failures are counted as errors in all conditions.

BERTScore F1 (Zhang et al., 2020) measures token-level semantic similarity between generated outputs and reference answers using contextual embeddings from RoBERTa-large. It serves as a secondary signal of output quality independent of exact-match accuracy.

4 Results

4.1 RQ1: Zero-Shot and Chain-of-Thought on MedQA

Table 1 summarises the performance of Meta-Llama-3-8B-Instruct on MedQA under zero-shot and CoT prompting.

On MedQA, zero-shot prompting outperforms CoT on every metric. The accuracy gap is modest but consistent (60.89% vs. 58.51%), and CoT also raises the parse failure rate from 0.8% to 3.6%.

Metric	Zero-Shot	CoT
Accuracy	60.89%	58.51%
Error Rate	39.6%	43.6%
Parse Failure Rate	0.8%	3.6%
BERTScore F1	0.8195	0.8144

Table 1: MedQA results for Llama-3-8B-Instruct (500 samples).

This suggests that for multiple-choice clinical reasoning, longer generated reasoning chains create additional opportunities for output instability and noisy intermediate steps.

However, the MedQA result does not imply that CoT is uniformly harmful. To test whether the effect generalizes across tasks, we also compare zero-shot and CoT prompting for BioMistral-7B on PubMedQA.

Metric	Zero-Shot	CoT
Accuracy	40.44%	49.56%
Error Rate	59.56%	50.44%
Parse Failure Rate	0.0%	0.0%
BERTScore F1	0.8021	0.8027

Table 2: PubMedQA zero-shot vs. CoT results for BioMistral-7B (450 samples).

For BioMistral-7B on PubMedQA, CoT improves accuracy from 40.44% to 49.56% and lowers error rate from 59.56% to 50.44%, with no parse failures in either condition and nearly identical BERTScore values. In other words, CoT helps on PubMedQA for this model, although it still remains below the corresponding RAG setup (56.19% accuracy). Taken together, these results suggest that the utility of CoT is task- and model-dependent rather than uniformly positive or negative.

4.2 RQ2: Retrieval-Augmented Generation on PubMedQA

We next evaluate retrieval-augmented generation on PubMedQA using BM25 top- $k=2$ retrieval. For BioMistral-7B, the corresponding zero-shot and CoT results were already reported in Table 2; Table 3 adds the RAG condition and reports the available RAG results for all three model families.

The updated results show that RAG is the strongest PubMedQA condition for both Llama-3-8B-Instruct and BioMistral-7B. For BioMistral-7B in particular, RAG improves on both zero-shot

Model	Acc.	Halluc.	Parse Fail	BERT
Llama-3-8B	73.39%	28.89%	3.11%	0.8500
BioMistral-7B	56.19%	45.56%	3.11%	0.8028
BioGPT	54.44%	89.11%	80.00%	0.8008

Table 3: PubMedQA RAG results with BM25 top-2 retrieval (450 samples per model).

(40.44%) and CoT (49.56%), reaching 56.19% accuracy while also reducing the reported hallucination rate to 45.56%. Llama-3-8B-Instruct achieves the best overall PubMedQA performance under RAG, with 73.39% accuracy and the lowest reported hallucination rate in this comparison. BioGPT remains the least reliable model: although its reported exact-match accuracy is 54.44%, it also has an 80.00% parse failure rate and an 89.11% hallucination rate, indicating severe instability in structured output.

4.3 RQ3: Effect of Fine-Tuning on Hallucination

We fine-tuned BioGPT on the full MedQA training split (10,178 samples) for 3 epochs. Training loss decreased steadily across epochs (2.05 $\rightarrow$ 1.73 $\rightarrow$ 1.63) with validation loss following the same trend (1.78 $\rightarrow$ 1.72 $\rightarrow$ 1.71), confirming convergence without overfitting. The fine-tuned model was evaluated on 500 MedQA test items alongside the zero-shot baseline. Table 4 reports the results.

Condition	Accuracy	Error Rate	Parse Fail
BioGPT zero-shot	23.44%	76.56%	23.20%
BioGPT fine-tuned	23.40%	76.60%	0.00%

Table 4: RQ3: BioGPT on MedQA before and after fine-tuning (500 evaluation samples).

Fine-tuning does not materially change BioGPT’s MedQA accuracy (23.44% $\rightarrow$ 23.40%), but it completely eliminates parse failures (23.20% $\rightarrow$ 0.00%). This suggests that the main benefit of SFT in our current setup is not better medical reasoning, but more reliable adherence to the required output format. Even after fine-tuning, BioGPT remains far below zero-shot Llama-3-8B-Instruct (60.89%), suggesting that model scale and instruction-tuning capability matter more than domain-specific fine-tuning alone for this task.

4.4 RQ4: RAG and Fine-Tuning Interaction

We applied RAG at inference time to the fine-tuned BioGPT model on PubMedQA to test whether the

two interventions are complementary. Since the SFT model was trained on MedQA and evaluated on PubMedQA, this is a cross-task transfer setting rather than a clean within-task comparison. Table 5 therefore focuses on the two PubMedQA conditions that can be compared directly: base BioGPT with RAG, and fine-tuned BioGPT with RAG.

Condition	Acc.	Halluc.	Parse Fail
BioGPT RAG only	54.44%	89.11%	80.00%
BioGPT SFT + RAG	41.61%	87.33%	69.56%

Table 5: RQ4: BioGPT under RAG and SFT+RAG conditions. Both conditions evaluated on PubMedQA (450 samples).

Compared with base BioGPT under RAG, SFT + RAG lowers the parse failure rate from 80.00% to 69.56% and slightly lowers the reported hallucination rate from 89.11% to 87.33%, but it also reduces exact-match accuracy from 54.44% to 41.61%. This mixed pattern suggests that fine-tuning partially improves output formatting under retrieval, but does not translate into better end-task QA performance.

An important methodological caveat must be noted: the SFT model was fine-tuned on MedQA (multiple-choice, A/B/C/D format) and then evaluated on PubMedQA (yes/no/maybe format). These are fundamentally different output formats. The observed degradation in the SFT + RAG condition therefore reflects not only BioGPT’s limited capacity, but also a *cross-task format mismatch* — the fine-tuned model learned to produce answer letters and lost the output flexibility required for yes/no/maybe responses. This confounding effect limits the conclusions that can be drawn about the RAG + fine-tuning interaction specifically.

Additionally, PubMedQA contains only 450 labeled samples in total, all of which were used for evaluation. This small dataset size limits statistical power and generalizability.

Despite these limitations, the result reveals a practically important finding: task-specific fine-tuning on a fixed output format changes how the model behaves under retrieval, but does not reliably improve PubMedQA performance. A rigorous evaluation of the RAG + fine-tuning interaction would require either (1) fine-tuning on PubMedQA specifically and evaluating with and without RAG on a held-out split, or (2) using a larger dataset that supports both training and evaluation splits. We leave this for future work.

4.5 Prediction Distribution Analysis

Output behavior varies sharply across models even when accuracy appears similar. On MedQA, CoT raises Llama-3-8B’s parse failure rate from 0.8% to 3.6%, showing that longer reasoning traces reduce format stability. By contrast, BioMistral-7B has zero parse failures under both zero-shot and CoT on PubMedQA. The clearest anomaly is BioGPT: under RAG it shows 54.44% exact-match accuracy, but also an 80.00% parse failure rate and 89.11% hallucination rate. After MedQA fine-tuning, SFT+RAG lowers parse failures to 69.56% but also lowers accuracy to 41.61%. These results show that answer accuracy alone is not enough to characterize reliability in medical QA; output-format stability is a separate and important failure mode.

5 Discussion

5.1 Effect of Model Scale

Across all conditions, model scale and instruction-tuning capability emerge as the dominant factors in hallucination reduction. Llama-3-8B-Instruct consistently outperforms both BioMistral-7B and BioGPT despite having no medical domain pre-training. This is consistent with recent findings that general large models with strong instruction tuning can match or exceed smaller specialized models on medical benchmarks (Singhal et al., 2023). Our results suggest that investing in larger, better-aligned general models may be more effective than domain-specific pretraining for reducing hallucination in medical QA.

5.2 Inference-Time vs. Training-Time Interventions

Our results reveal an important asymmetry between inference-time and training-time interventions. RAG substantially reduces hallucination for Llama-3-8B and also outperforms both zero-shot and CoT on PubMedQA for BioMistral-7B, but it actively hurts BioGPT. This suggests that RAG is only effective when the model has sufficient instruction-following capacity to leverage retrieved context. Fine-tuning improves BioGPT format consistency by eliminating parse failures on MedQA, but it does not improve BioGPT’s answer accuracy. However, the absolute accuracy of the fine-tuned model (23.40%) remains far below zero-shot Llama-3-8B (60.89%), suggesting that model

scale and instruction-tuning capability matter more than fine-tuning alone for hallucination reduction.

5.3 Format Rigidity as a Hidden Risk of Fine-Tuning

A particularly important finding emerges from our RQ4 experiment. BioGPT struggles on PubMedQA even after retrieval augmentation. In our cross-task transfer setting, applying MedQA fine-tuning before RAG does not improve end-task accuracy, although it does reduce parse failures relative to base BioGPT under RAG. This indicates that format adaptation and task performance can move in different directions.

This finding has practical implications beyond our specific experimental setup: fine-tuning a model on a fixed output format introduces format rigidity that may prevent it from generalizing to other tasks at inference time, and can compound existing model weaknesses rather than mitigate them. Practitioners deploying fine-tuned models in multi-task clinical settings should carefully evaluate format compatibility before combining fine-tuning with retrieval augmentation.

We acknowledge an important caveat: our RQ4 setup involves a cross-task format mismatch, and PubMedQA’s 450 labeled samples are too few to support both a fine-tuning split and a reliable evaluation split. A fully controlled interaction test — fine-tuning and RAG evaluation within the same format and dataset — remains for future work. Nevertheless, the finding that SFT changes output formatting more than end-task accuracy stands independently of this limitation.

6 Conclusion

We presented a controlled empirical study of hallucination behavior in medical LLMs across four intervention conditions. Our key findings are: (1) larger instruction-tuned models hallucinate less under RAG, with Llama-3-8B achieving 73.39% accuracy on PubMedQA; (2) CoT prompting has mixed effects, lowering performance on MedQA with Llama-3-8B-Instruct but improving over zero-shot on PubMedQA with BioMistral-7B; (3) fine-tuning BioGPT primarily improves output format reliability rather than answer accuracy on MedQA; and (4) combining fine-tuning with RAG changes BioGPT’s behavior on PubMedQA, but does not yield a clean performance gain. These findings highlight that model scale and instruction-tuning

capability are more important than domain-specific adaptation for hallucination reduction, and that combined interventions must be evaluated for format compatibility rather than assumed to be additive.

7 Limitations

Several limitations of this work should be noted. First, our RQ4 evaluation involves a cross-task format mismatch: BioGPT was fine-tuned on MedQA (multiple-choice, A/B/C/D) but evaluated with RAG on PubMedQA (yes/no/maybe). A rigorous interaction test would require fine-tuning and evaluation within the same format and dataset. PubMedQA’s 450 labeled samples are too few to support both fine-tuning and a reliable evaluation split, which is why we could not conduct this experiment within the current study.

Second, our fine-tuning used only 3 epochs with standard SFT; longer training or alternative objectives such as RLHF or DPO may yield greater hallucination reductions. Finally, denser retrievers than BM25 top-2 may improve RAG quality and faithfulness.

8 Author Contributions

All authors contributed to both the technical implementation and writing process. **Aidan Perez** contributed baseline evaluation development, dependency and build fixes, project documentation updates, and core manuscript integration. **Dang Dinh** contributed RAG evaluation implementation, metrics refinement, references updates, and major results/discussion revisions (including RQ3 and RQ4 updates). **Kaveri Goel** contributed experiment-pipeline engineering, including zero-shot/CoT/RAG workflow support, batching inference, retrieval hyperparameterization (e.g., top- k), and associated debugging fixes. **Ying Huang** contributed substantial manuscript development and revision (including CoT, related work, results, and future-work framing), as well as project coordination and final consolidation. All authors reviewed and approved the final manuscript.

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